

between ChatGPT and Cohere. Despite solid performance, their method does not require fine-tuning.

Team L3i++_{STB:6} (Tran et al., 2024) conduct a comparative study among three groups of methods: metric-based models, fine-tuned classification language models (RoBERTa, XLM-R), and a fine-tuned LLM, LLaMA-2-7b. They find LLaMA-2 outperforming the methods from the other groups in MGT detection. The team reveals the analysis of errors and various factors in their paper.

Team MLab_{STB:7} (Li et al., 2024) fine-tune DeBERTa and analyze the embeddings from the last layer. They provide insights into the embedding space of the model.

Team Werkzeug_{STB:8} (Wu et al., 2024) utilizes RoBERTa-large and XLM-RoBERTa-large to encode the text. They tackle the problem of anisotropy in text embeddings produced by pre-trained language models (PLMs) by introducing a learnable parametric whitening (PW) transformation. Furthermore, to capture the features of LLM-generated text from different perspectives, they use multiple PW transformation layers as experts under the mixture-of-experts (MoE) architecture equipped with a gating router in their final solution.

Team TrustAI_{STB:14} (Urlana et al., 2024) explore different pretrained and statistical models for detecting synthetic text, ultimately selecting the RoBERTa-base OpenAI Detector for its effectiveness. This model, originally fine-tuned with outputs from the 1.5B-parameter GPT-2 model, is further fine-tuned on the Subtask-B dataset.

Team MGTD4ADL_{STB:17} (Chen et al., 2024) combine traditional Transformer models (RoBERTa-base, RoBERTa-large, GPT-2-small, XLNet, T5-small) with Sentence Transformers(all-mpnet-base-v2 and all-roberta-large-v1). They further diversify their approach by leveraging different data augmentation techniques and experimenting with various loss functions such as Cross-Entropy (CE), Supervised Contrastive Learning (SCL), and Dual Contrastive Loss (DUALCL).

Team scalar_{STB:18} (M et al., 2024) employ an ensemble of three RoBERTa-base models using an individual validation set for each model.

Team UMUTeam₂₃ (pan et al., 2024) use fine-tuned RoBERTa model combined with syntactic features of the text such as word length, part of speech, function word frequency, stop-word ratio, and sentence length.

Team QUST_{STB:36} (Xu et al., 2024) use fine-tuned RoBERTa and DeBERTa models, integrating them through a stacking ensemble technique.

Team MasonTigers_{STB:38} (Puspo et al., 2024) implement an ensemble of 3 PLMs: RoBERTa, DeBERTa, and ELECTRA. Additionally, they employ zero-shot prompting and use a fine-tuned FLAN-T5 model.

Team RUG-5_{STB:41} (Darwinkel et al., 2024) expands the architecture of DistilBERT models by adding an additional classification layer that incorporates 20 linguistic-stylistic features. They also explore the use of Random Forest classifier on top of embeddings from DistilBERT combined with the same set of linguistic-stylistic features.

Team RUG-D_{STB:44} (Brekhof et al., 2024) focus on fine-tuning DeBERTa models.

Team Groningen-F_{STB:45} (Donker et al., 2024) trained traditional machine learning models (SVM and FFNN) with features including tense of the sentence, the voice of the sentence, the sentiment of the sentence, and the number of pronouns vs. proper nouns.

Team DUTh_{STB:49} (Kyriakou et al., 2024) explore traditional machine learning algorithms along with BERT for their task. Ultimately, they proceed with BERT fine-tuned for 5 epochs.

Team AT_{STB:58} (Wei, 2024) adopts three different semantic embedding algorithms, GLOVE, n-gram embeddings and SentenceBERT as well as their concatenation to identify the generator in Subtask B. The author finds that these pre-trained embeddings, while fast to compute, are not as effective as a fine-tuned RoBERTa model.

Team iimasNLP_{STB:61} (Valdez et al., 2024) fine-tune 4 different language models to classify text generated by different models: ERNIE, SpanBERT, ConvBERT and XLNet. They find out that RoBERTa is a stronger classifier. In general this shows how fine-tuning PLMs is an effective approach to identify the generator model.

Team CLULab-UofA_{STB:62} (Rezaei et al., 2024) combine LLM fine-tuning with contrastive learning, specifically using triplet loss.

A.4 Boundary Identification

Team TM-TREK_{STC:1} (Qu and Meng, 2024) achieved the highest performance in Subtask C by employing an ensemble of models including Longformer, Bigbird, and XLNet for long-text sequence labeling. A simple voting mechanism was used to aggregate the output logits. Their innovative strategy also involved integrating LSTM and CRF layers atop various pre-trained language models (PLMs), along with continuous pretraining, fine-tuning, and utilizing dice loss functions to enhance model performance.

Team AIpom_{STC:2} (Shirnin et al., 2024) introduced a two-stage pipeline that combines outputs from an instruction-tuned, decoder-only model (Mistral-7B-OpenOrca) with two encoder-only sequence taggers. Initially, they trained an instruction-tuned autoregressive model to insert a [BREAK] token into input texts, delineating human-written parts from machine-generated ones. Subsequently, these annotated texts were processed by an encoder-based model for sequence tagging, differentiating human-written tokens (0) from machine-generated tokens (1). An additional encoder trained on a blend of raw and annotated texts further refined sequence tagging. The average change point positions predicted by both encoders served as the final boundary estimation.

Team USTC-BUPT_{STC:3} (Guo et al., 2024) approached the task as a token classification challenge, opting to fine-tune a DeBERTa model enhanced by data augmentation techniques derived from the training set. They reported that DeBERTa-base outperformed other models, and explored the efficacy of sequence labeling (e.g., BIOS) in detecting boundaries within mixed texts. The potential of various layers, including CRF and Dropout, was also examined for their impact on system performance.

Team RKadiyala_{STC:6} (Kadiyala, 2024) focused on fine-tuning various encoder-based models appended with a Conditional Random Field (CRF) layer, noting that DeBERTa-V3 yielded the best results on the development set.

Team DeepPavlov_{STC:7} (Voznyuk and Konovalov, 2024) fine-tuned the DeBERTa-v3 model using a specially prepared dataset with augmented texts, created by modifying prefixes and suffixes of original texts. They emphasized the importance of augmented data quality in achieving a mean absolute error (MAE) of 15.20903.

Team RUG-5_{STC:17} (Darwinkel et al., 2024) utilized an augmented Longformer model, incorporating extra features into the output state of each token to enrich them with contextual information. This approach aimed at improving token-level classification by leveraging linguistic-stylistic features beyond simple PLM optimization.

Team TueCICL_{STC:22} (Stuhlinger and Winkler, 2024) experimented with character-level LSTMs and LSTMs using pretrained Word2Vec embeddings, demonstrating that smaller models could compete with transformer models in the boundary detection task.

Team jelarson_{STC:25} (Larson and Tyers, 2024) explored rule-based methods and linear regression techniques, identifying specific patterns in the training data that could inform better data collection practices, such as ensuring a more randomized and unbiased dataset.

Team TueSents_{STC:26} (Pickard and Do, 2024) extracted textual features at the sentence level using tools like SpaCy and trained a lightweight BiLSTM model for boundary prediction, achieving an accuracy of 0.7 and MAE of less than 0.5 on the development set.

Team MasonTigers_{STC:27} (Puspo et al., 2024) combined TF-IDF, PPMI, and RoBERTa features with linear regression and Elastic Net, culminating in an ensemble approach based on a weighted development set.

Team Unibuc-NLP_{STC:28} (Marchitan et al., 2024) framed the task as a token classification problem, merging character-level features (extracted via CNN) and word embeddings within a BiLSTM model, further exploring the addition of CRF for enhanced performance.